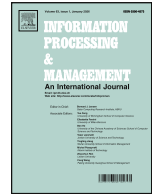




Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/ipm

LLM-assisted fake news detection with adaptive boosting framework incorporating contrastive learning

Shu Yin^{a,b}, Yuchen Wang^b, Dongpeng Hou^c, Wenxin An^c, Chao Gao^b,
Xianghua Li^b, Zhen Wang^{d,*}^a School of Computer Science, Northwestern Polytechnical University, Xi'an, 710072, Shaanxi, China^b School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, 710072, Shaanxi, China^c School of Mechanical Engineering, Northwestern Polytechnical University, Xi'an, 710072, Shaanxi, China^d School of Cybersecurity, Northwestern Polytechnical University, Xi'an, 710072, Shaanxi, China

ARTICLE INFO

Keywords:

Fake news detection
Social network
Large language model
Text analysis

ABSTRACT

With the rapid development of digital technologies, the widespread spread of fake news on social media platforms may threaten social stability. Existing methods often use small models as feature extractors for news content, relying on the inherent pattern differences between true and false news as the basis for fake news detection. However, smaller models cannot perceive latent background information in news content. Additionally, fake news is much less prevalent than true news on real social platforms, which may lead to issues with data collection imbalance of true and false labels. These challenges may limit the effectiveness of existing detection methods. Therefore, we propose a Large Language Model-assisted Fake News Detection method with Adaptive Boosting, denoted as LFND-AB. Specifically, we first employ the large language model to generate analyses of the truthfulness and falseness of news text, which contains background information about the news. We then design an explainable base learner for fake news detection that utilizes contrastive learning to align the similarity between the news text and the corresponding generated analyses to obtain higher-quality fused representations. Furthermore, we design an adaptive ensemble boosting framework based on the base learner. This framework adjusts the weights of misclassified samples to improve the detection performance in the next base learner, thereby optimizing the model's ability to handle label imbalance. The comprehensive experiments conducted on benchmark datasets demonstrate that both the proposed base learner and LFND-AB outperform State-Of-The-Art (SOTA) methods. The implementation code is available at <https://github.com/cgao-comp/FEND>.

1. Introduction

Social media is the product of the deep integration of the Internet and traditional social pathways (Hou et al., 2024). As digitization continues to develop, the various social platforms have revolutionized the way of communication and information acquisition. These digital changes have not only brought convenience to our daily lives but also caused an increase of fake news on social platforms (Dou et al., 2021). In the real world, the growing scale of fake news poses severe challenges to individual decision-making and public trust,

* Corresponding author.

E-mail addresses: yinshu@mail.nwpu.edu.cn (S. Yin), wany810@mail.nwpu.edu.cn (Y. Wang), hdp0918@mail.nwpu.edu.cn (D. Hou), wenxinan@mail.nwpu.edu.cn (W. An), cgao@nwpu.edu.cn (C. Gao), li_xianghua@nwpu.edu.cn (X. Li), w-zhen@nwpu.edu.cn (Z. Wang).<https://doi.org/10.1016/j.ipm.2026.104789>

Received 22 September 2025; Received in revised form 31 January 2026; Accepted 30 March 2026

Available online 15 April 2026

0306-4573/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

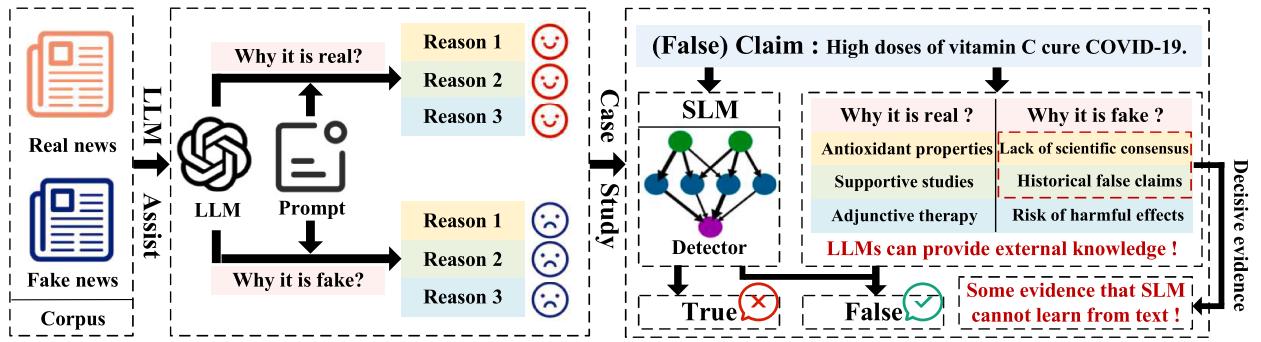


Fig. 1. The illustration of the potential of Large Language Models (LLMs) for fake news detection task. LLMs could capture external knowledge, such as historical facts, that Small Language Model (SLM) based methods fail to learn from training datasets. Moreover, LLM's expert-level inference capability can significantly assist in fake news detection.

with the potential to influence public opinion and exacerbate social instability (Wu et al., 2021a). Moreover, existing studies show that fake news spreads faster and more widely than real news (Vosoughi et al., 2018). Without timely detection and clarification, fake news may result in long-term public misperception. Therefore, investigating how to efficiently and accurately verify large-scale online misinformation not only holds academic value but also has significant practical implications (Hu et al., 2023). Such research can inform the development of deployable tools for integration into social media platforms, news organizations, and governmental monitoring systems, playing a crucial role in preserving information authenticity in online environments.

Some studies leverage social context signals (e.g., user comments, publisher profiles, and interaction patterns) during news propagation to assist in detecting fake news (Cui & Jia, 2024). Graphs are widely used in modeling the news dissemination process, where nodes represent news articles or comments, and edges represent relationships through comments or shares (Wu & Hooi, 2023). This method helps in analyzing the spread patterns and interactions within the network, effectively highlighting the characteristic patterns of fake news. For example, Bian et al. (2020) construct a bidirectional graph for the news propagation graph to capture the characteristics of rumor propagation and diffusion. Hyper-ISDR uses random hypergraph theory to more reasonably model the spread of Internet rumors and deeply explores the impact of group interaction on the dynamics of rumor propagation (Xia et al., 2025). However, these methods typically rely on information that is costly to obtain (e.g., via large-scale web crawling) and often require time to accumulate (Wu et al., 2023b), which may be limited in certain scenarios (such as when comment data is sparse or during the early stages of news propagation). As a result, some research focuses solely on the news content for reality verification, referred to as content-based fake news detection. Specifically, considering that fake news is often deliberately crafted to attract public attention, content-based methods detect fake news by extracting semantic and emotional features from the textual content (Zhang et al., 2021). However, news articles often contain latent background knowledge and cues that are difficult for conventional Small Language Models (SLMs) to capture effectively, limiting the depth of understanding these models can provide.

As a powerful tool, Large Language Models (LLMs) can capture and analyze fine-grained segments (such as historical factual contexts) based on extensive knowledge bases and inference capabilities (Chen & Shu, 2023; Wang et al., 2024b). Although LLMs can mine the content within news articles, current studies indicate that directly using them as fake news classifiers is suboptimal (Hu et al., 2024). Therefore, how to effectively use LLMs to enhance the understanding of news content poses a significant issue. Furthermore, the proportion of verifiable fake news is small compared to the real news on the Internet (Lu et al., 2024; Sheng et al., 2022), and the acquisition and cleaning of data require a lot of human resources. Therefore, designing a robust model to detect fake news effectively under label imbalance conditions is crucial.

To address these issues, we propose an LLM-assisted Fake News Detection method with Adaptive Boosting (LFND-AB) using only textual content from the news, which includes textual content analysis, fake news base learner design, and ensemble boosting strategy. In the textual analysis phase, leveraging the extensive knowledge base and expert-level inference capabilities of LLMs, we generate reasons for each piece of news to analyze and support its truthfulness and falseness as Fig. 1. In the model design phase, an explainable base learner is designed to efficiently utilize these reasons and the original text to identify the veracity of the news. More specifically, after obtaining the embeddings of the textual content and its associated reasons using the Pre-trained BERT, we then propose a content-reason interaction attention module, which aggregates and reweights the textual content and each reason. Then, a sampling-based contrastive learning module is developed to improve the embedding quality of original textual content. Finally, a composite loss function is designed to differentiate the embedding of reasons supporting truthfulness or falseness, and guide the model for training. While the above LLM-assisted base learner can already perform well on its own, it may still be susceptible to label imbalance. In the ensemble boosting strategy, to further enhance robustness, we design an adaptive boosting framework to dynamically adjust each sample weight in the next base learner. Specifically, we reduce the weights of the samples that are correctly classified in the previous round and increase the weights of the samples that are misclassified. This ensemble mechanism allows the model to focus on harder cases in subsequent rounds, improving its overall effectiveness. A series of experiments demonstrates the effectiveness of the proposed LFND-AB method and the base learner. In summary, this paper makes the following contributions:

- LLMs are used to generate reasons supporting the truthfulness and falseness of news, providing external background knowledge to assist in fake news detection.
- An explainable fake news base learner is designed to integrate textual content and reasons, enhancing embedding quality dynamically.
- An adaptive boosting framework is introduced, specifically tailored for label-imbalanced datasets in fake news detection.

2. Related work

2.1. SLM-Based fake news detection

As the fake news in social networks increases rapidly, automatically detecting fake news has become an urgent issue (Zhou & Zafarani, 2020). The task of detecting fake news can be seen as a classification task for news. According to the features used, fake news detection methods can be divided into two categories: Social context-based methods and news content-based methods.

Context-based methods use social context, such as user characteristics (Dou et al., 2021), comments (Jin et al., 2016), propagation networks (Zhang et al., 2024), to provide additional information to help detect fake news. Ma et al. (2018) model the information dissemination process as a tree with both bottom-up and top-down directions, using recurrent neural networks designed for tree structures to identify rumors. Due to the superior ability of GNNs to represent structured data, Bian et al. (2020) propose a bidirectional graph convolutional network that learns the characteristics of rumor propagation in both bottom-up and top-down directions, thereby enhancing the accuracy of rumor detection. To better capture structured information, GLAN constructs a heterogeneous graph for each post by establishing relationships between the original posts, reposts, and users, fully leveraging the correlations between different types of nodes for rumor detection (Yuan et al., 2019). Additionally, considering the poor robustness of existing methods to noisy propagation structures, Sun et al. introduce graph adversarial contrastive learning to handle such complex scenarios, where contrastive learning is incorporated as part of the loss function to perceive the differences between similar and different categories (Sun et al., 2022). Bazmi et al. (2023) introduce a multi-view attention network, which captures semantic and propagation structure clues through designed attention modules, providing reasonable explanations for fake news detection. Moreover, the historical information of the publisher can also assist in detecting fake news. For instance, Shu et al. (2019) study the relationship between user profiles and fake news, proposing a method to represent implicit features of user profiles, demonstrating the potential of user features in distinguishing fake news. However, acquiring the additional information required by these methods involves significant resources, reducing their practicality.

Content-based fake news detection has garnered widespread attention due to the relatively low cost of dataset acquisition (Shu et al., 2017). These methods typically examine textual patterns in news articles, including writing style and article inclination (Przybyla, 2020), semantic features (Qi et al., 2021), and emotional characteristics (Luvembe et al., 2023). And most of the corresponding researches use small models to extract content features (Capuano et al., 2023). For example, Chen et al. (2018) use the Recurrent Neural Network (RNN) to capture long-term dependent features from news text sequences, while introducing the soft attention mechanism to capture time-varying features. To exploit the emotional information in the text, Zhang et al. (2021) propose the dual sentiment feature for fake news detection, which extracts the publisher sentiment and social sentiment and the similarities and differences between them from the text to judge the authenticity of the news. Guo et al. (2023) construct positive and negative evidence graphs using news signals from the dataset. When detecting a particular news article, it is compared with the evidence graphs to generate relevant matching vectors to assist in fake news detection. To address fake news detection across multiple domains, Zhu et al. (2022b) introduce a multi-view fake news detection framework. This framework models news from three perspectives: semantics, sentiment, and style. Then it uses the unique information from these domains to effectively identify fake news. Learning from multiple views, the framework captures both differences and commonalities across domains, improving the accuracy of fake news detection. To further leverage auxiliary information, multimodal methods have been proposed to jointly utilize text and image features for fake news detection. For example, Wu et al. (2021b) propose the Multimodal Co-Attention Network (MCAN), which models the inter-modal interactions between visual and textual content, addressing the limitations of simple feature concatenation. Furthermore, CMMTN enhances multimodal fake news detection accuracy by employing a masked transformer architecture to fuse cross-modal signals and dynamically filter out irrelevant inter-modal context (Wang et al., 2023). Considering that fake news usually carries specific negative intentions, Wang et al. (2024a) construct a hierarchical intent classification system based on psychological theory, and gradually infer the potential intentions of the article through the encoder-decoder model to detect fake news. Different from traditional methods that treat news-related text data as sequences, Wu et al. (2023a) model them as graph-structured data and captures long-range semantic dependencies through neighborhood propagation, thereby effectively reducing information redundancy and improving the accuracy of semantic representation. However, news often contains rich background information and clues that are difficult for small models to extract. At the same time, the knowledge that pre-trained models learn during their pre-training phase limits the model's capabilities and further restricts the performance of future fake news detection. For example, Bidirectional Encoder Representation from Transformers (BERT) is pre-trained on corpora like Wikipedia, making it difficult to handle the rich background knowledge and potential clues in future news (Sheng et al., 2021). To address these issues, we propose the introduction of LLMs to understand the news content and generate analytical statements for each news instance, which incorporate additional information beyond the original text.

2.2. LLM-Based fake news detection

Recently, LLMs have demonstrated exceptional abilities in various fields, particularly in classification, reasoning tasks, and other classic natural language processing tasks (Chang et al., 2024; Ouyang et al., 2022). In the specific context of fake news detection, Kuntur et al. (2024) provide a survey of LLM-based fake news detection methods, outlining how they extend pre-LLM approaches and summarizing their advantages, limitations, and open challenges. Liang et al. (2024) explore a multi-agent debate framework, showcasing the collaborative problem-solving abilities of LLM agents. Strong et al. (2024) present a zero-shot fact verification method based on natural logic and LLMs, emphasizing the model's generalization ability and interpretability. However, LLMs have not performed well as fake news detectors and come with high training costs (Hu et al., 2024). With their powerful reasoning capabilities, designing reasonable prompts for samples and analyzing the outputs to indirectly improve detection ability has shown potential for LLMs in the fake news detection domain. Li et al. (2024) propose a framework that uses externally acquired knowledge as prompts for LLMs, achieving good results in a zero-shot environment. Cheung and Lam (2023) train a detector based on LLaMa using external knowledge. Furthermore, Wan et al. (2024) integrate LLMs to generate diverse user reactions, execute explainable agent tasks, and merge expert predictions, thus improving detection accuracy in the fake news detection process. In parallel, Courtroom-FND formulates fake news detection as a multi-role courtroom debate with a judge and two adversarial agents that switch roles, aiming to reduce cognitive bias and improve decision reliability (Jin et al., 2025b). Moreover, PMTL-DisCo and CAPE-FND investigate prompting-based few-shot and zero-shot fake news detection (Jin et al., 2025a,c). SheepDog differs from previous studies that use LLM to make accuracy judgments to distinguish false information from true information (Wu et al., 2024). Instead, it studies how LLM can enhance the style robustness of fake news detectors, especially by injecting style-diverse reconstructions and content-centric cues during training. However, there are challenges associated with using LLMs for fake news detection. One key issue is that the content generated by LLMs may sometimes suffer from low quality, which can degrade the overall performance. Therefore, designing a dynamic interaction attention module that selectively focuses on high-quality, relevant information from the LLM-generated content is crucial. What's more, verifiable fake news is much less frequent than real news on actual social networks. Thus, improving the detection performance under imbalanced data conditions is an important consideration. While recent methods such as ARG (Hu et al., 2024) and SheepDog (Wu et al., 2024) also leverage LLM-generated rationales, our approach introduces several novel designs. First, unlike ARG, which only generates single-perspective rationales supporting truthfulness, we employ dual-perspective prompting to generate both supporting and opposing reasons, providing more balanced and comprehensive interpretive signals. Second, we design a contrastive learning module that encourages the model to distinguish between aligned and misaligned news-reason pairs. Third, we further integrate these dual-perspective reasons into a boosting-based ensemble framework, where each base learner is adaptively trained with reweighted examples to handle data imbalance. These components together enable our model not only to benefit from LLM knowledge but also to enhance robustness and explainability under limited supervision.

3. Method

Fake news detection is a binary classification task, where the core objective is to identify the veracity of news articles (true or false). Given a news item with its textual content denoted as C , our approach initiates by generating two distinct sets of reasons by LLMs: $R = \{r_1, r_2, r_3\}$ supporting the news as true, and $F = \{f_1, f_2, f_3\}$ suggesting it is false. Using the original text C along with these generated reasons, the main goal is to develop a model function:

$$f(C, F, R) \rightarrow y, \quad (1)$$

where y is the authentic label indicating the truthfulness of the news. The proposed LFND-AB includes three main phases: the LLM assistance phase for analytical data generation, the base learner design phase for fake news detection, and the ensemble strategy phase for performance boosting. Each phase is presented in the subsequent parts.

3.1. Textual analysis and reason data generation

Considering the high overhead of multimodal information collection, our method focuses solely on the traditional textual content C of the news. As extensive knowledge bases and expert systems, LLMs have advanced reasoning capabilities and contextual comprehension by analyzing and generating insights that stand expert perspective. Leveraging these properties, LLMs are employed to mine the textual content C , focusing not only on contextual information but also on views that smaller models cannot deal with, such as historical and factual contexts, emotional tones, etc.

The rationale for using LLMs instead of traditional external knowledge sources, such as fact-checking databases or structured knowledge graphs, lies in the scalability and flexibility that LLMs provide. While traditional knowledge sources are accurate and manually verified, they often demand significant maintenance efforts and are limited in scope (Wang et al., 2024b; Xie et al., 2024). In contrast, LLMs can offer a more dynamic and adaptable solution (Pan et al., 2024). However, although LLMs enhance the background knowledge associated with news content, previous studies have shown that they are not ideally suited to function as standalone fake news detectors. Instead, they perform more effectively when used to guide smaller models in their decision-making processes. Therefore, our method does not rely solely on the reasons generated by LLMs as the foundation for detecting fake news. Instead, we integrate these generated reasons with the original news content. We introduce a dual-perspective prompting strategy that guides LLMs to generate both true and false reasoning, offering richer evidence than ARG. Specifically, for each news text C , our methodology utilizes an LLM to generate three analytical statements supporting the news as true ($R = \{r_1, r_2, r_3\}$) and three contesting it as false

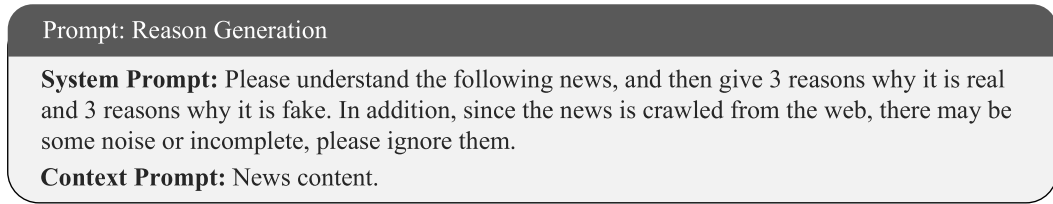


Fig. 2. Prompts used for generating reasons in the fake news detection process. The system prompt requests the generation of 3 reasons supporting the authenticity and 3 reasons supporting the falseness of the news, while the context prompt provides the news content for analysis.

($F = \{f_1, f_2, f_3\}$). ChatGPT-4.0¹ is used as the LLM for generating reasons, and prompts are shown in Fig. 2. This balanced approach of generating reasons for both truthfulness and falseness ensures a comprehensive dataset, facilitating a more nuanced understanding and training of the subsequent fake news detection model.

In our experiments, generating reasons for each Chinese news item takes an average of 3.4 seconds with 208 input tokens and 242 output tokens, while each English item takes an average of 3.6 seconds with 297 input tokens and 298 output tokens. For the Chinese dataset (1,000 news items), the input and output costs are \$6.24 and \$14.52, respectively, resulting in a total of \$20.76. For the English dataset (1,000 news items), the input and output costs are \$8.91 and \$17.88, respectively, with a total of \$26.79.² To reduce latency and computational cost, the generation of truthfulness and falseness reasons via LLMs is performed as a one-time preprocessing step for each news article. Rather than querying the LLM repeatedly during training or inference, the generated reasons are stored in structured JavaScript Object Notation (JSON) files, indexed by their corresponding news IDs. This design explicitly separates the high-cost LLM reasoning stage from the online detection stage, ensuring that the inference latency is bounded by model computation rather than API calls. Furthermore, the preprocessing stage can be executed offline or in batch mode, which further improves the engineering feasibility of the proposed framework under different system constraints. In real-world applications, when a new news article is encountered for detection, we can use the above prompt strategy to generate corresponding reasons, then save the reasons in the JSON file for downstream use. Moreover, the weight of the detection model should be updated periodically through fine-tuning with new labeled data, and the saved generated reason files should be updated when shifts in topic or language style are observed.

3.2. Fake news base learner design

Following the generation of datasets via LLM, our objective is to use the synthesized reasons R and F effectively to determine the veracity of news articles. To achieve this, an explainable model is proposed to obtain composite representations for news, which is also the base learner of the ensemble boosting framework. The model includes three core modules: the content-reason interaction attention module, the contrastive learning module, and the loss function. Fig. 3 shows the framework of the model.

3.2.1. Interaction attention module

Although LLMs can generate the truthfulness or falseness reasons based on external knowledge, their inherent hallucination problem may lead to fabricated evidence. This undermines the feasibility of using LLMs as direct predictors for fake news detection (Hu et al., 2024). The valid generated reasons often contain background knowledge related to the news content, which provide significant value for the model to better understand the veracity of the news. To address this, we leverage the attention mechanism to implicitly learn the dynamic scores of different reasons based on the downstream ground-truth labels. This allows the model to assign higher weights to reliable reasons while down-weighting less trustworthy reasons. In this way, the attention mechanism acts as a filter that mitigates the unreliable generated reasons in a data-driven manner.

The interaction attention module dynamically focuses on the original news and each pair of reasons, which includes the reason aggregation strategy and explainable weight adjustment strategy. The weight coefficient reflects the importance of each reason, which can assist in judging fake news while maintaining explainability and improving the effectiveness and transparency of the fake news detection process. While our model adaptively weights each reason via interaction attention, it does not explicitly assess causal relationships or inter-reason redundancy, which we consider these aspects as important directions for future research.

Given the news text C and the sets of reasons $R = \{r_1, r_2, r_3\}$ for truthfulness and $F = \{f_1, f_2, f_3\}$ for falseness, we employ the pre-trained BERT model to encode these texts into embeddings as:

$$\begin{cases} h_C = \text{BERT}(C) \\ h_{r_n} = \text{BERT}(r_n) \\ h_{f_n} = \text{BERT}(f_n) \end{cases}, \quad n = 1, 2, 3, \quad (2)$$

where h_C , h_{r_n} , and h_{f_n} represent the embeddings of the main content C , reasons r_n supporting truthfulness, and reasons f_n supporting falseness, respectively. Subsequently, a content-based interaction attention mechanism is developed to dynamically focus on each

¹ <https://openai.com/>

² Using ChatGPT-3.5-turbo as a more economical alternative, the input and output costs for the Chinese dataset (1,000 items) are \$0.08 and \$0.35. For the English dataset, the corresponding input and output costs are \$0.10 and \$0.14.

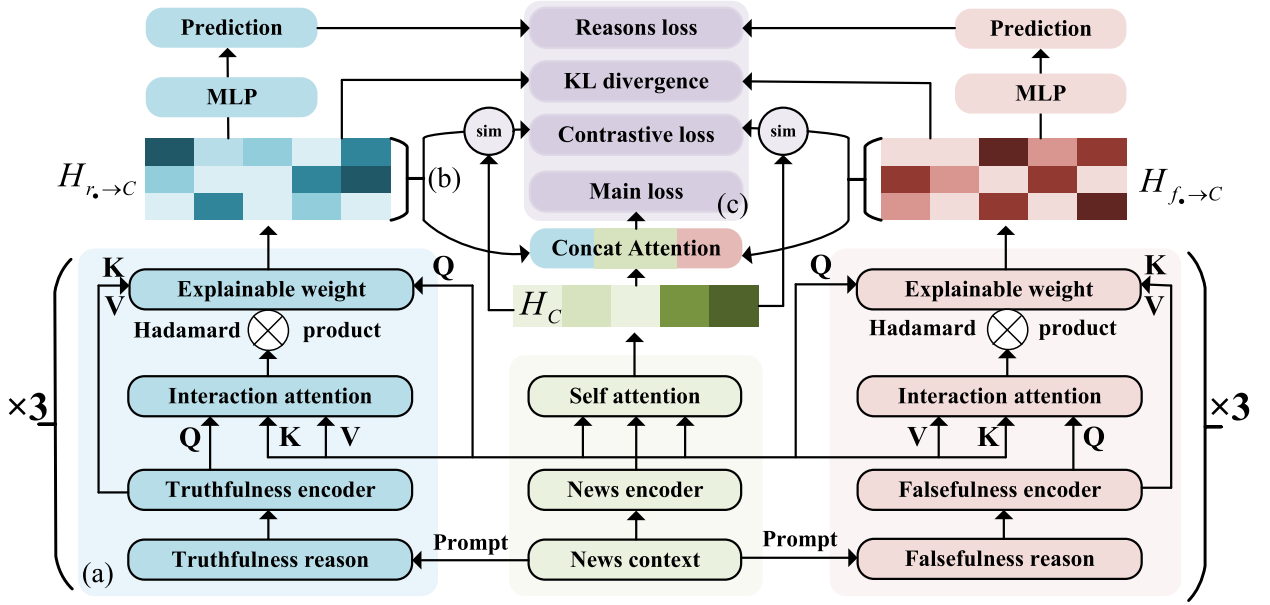


Fig. 3. The illustration of an explainable base learner for fake news detection. (a) is the content-reason interaction attention module to obtain the information aggregation between the textual content and reasons, and the weight of each content-reason pair is evaluated. (b) is the contrastive learning module based on the self-attention mechanism, enhancing the embedding quality of the original content. (c) is the joint loss to differentiate reason embeddings representing opposing stances on truthfulness and falseness.

pair between the original text and each reason, which includes the reason aggregation strategy and explainable weight adjustment strategy. The reason aggregation strategy for the textual content is shown as follows:

$$H_{X_n \rightarrow C} = \text{softmax} \left(\mathbf{Q}(h_{X_n}) \cdot \mathbf{K}(h_C)^T / \sqrt{d} \right) \mathbf{V}(h_C), \quad (3)$$

where $X_n \in X$, and $X = R \cup F = \{r_1, r_2, r_3, f_1, f_2, f_3\}$. $\mathbf{Q}(h_{X_n}) = W_Q h_{X_n}$, $\mathbf{K}(h_C) = W_K h_C$, and $\mathbf{V}(h_C) = W_V h_C$. d is the dimensionality. $H_{X_n \rightarrow C}$ denotes the n -th reason-guided content embedding. After using Eq. (3), six distinct content embeddings that aggregate reasons supporting truthfulness or falseness are obtained, separately. Further, the importance of six distinct embeddings $H_{X_n \rightarrow C}$ are different intuitively. To evaluate their importance, an explainable weight coefficient³ $W_{C \rightarrow X_n}$ is introduced using a reverse attention mechanism from the textual content to each reason, measuring how useful the n -th reason is for judging the veracity of the current news item.

$$W_{C \rightarrow X_n} = \text{MLP}(\text{softmax} \left(\mathbf{Q}(h_C) \cdot \mathbf{K}(H_{X_n})^T / \sqrt{d} \right) \mathbf{V}(H_{X_n})), \quad (4)$$

where MLP transfers the attention vector to a weight number. Then the explainable coefficient $W_{C \rightarrow X_n}$ can adjust the weight of the reason-attention based textual embedding $H_{X_n \rightarrow C}$.

$$H_{X_n \rightarrow C} \leftarrow W_{C \rightarrow X_n} \cdot H_{X_n \rightarrow C}. \quad (5)$$

3.2.2. Contrastive learning module

Six content embeddings incorporating true or false reasons are dynamically reweighted after executing the content-reason interaction attention module. Contrastive learning is a representation learning paradigm that aims to learn discriminative features by bringing semantically similar samples closer and pushing dissimilar samples apart in the latent space (Wang et al., 2022). Since it has shown strong capability in improving representation learning and has achieved promising results across various tasks (Hu et al., 2025), a contrastive learning strategy is employed to enhance the representation quality of original textual content. Based on the ground-truth labels, the embeddings of the original content are contrasted with those of the corresponding reasons by minimizing the distance between positive similar pairs (i.e., content and matching reason) and maximizing the distance between negative dissimilar pairs (i.e., content and opposing reason). To implement the procedure, we first utilize a self-attention mechanism to enhance the textual relationships within the original content. The computation is defined as follows:

$$H_C = \text{softmax} \left(\mathbf{Q}(h_C) \cdot \mathbf{K}(h_C)^T / \sqrt{d} \right) \mathbf{V}(h_C). \quad (6)$$

³ These weight coefficients for each reason can be ranked, and the explainable weights are demonstrated in the Case Study.

After the attention mechanism, the contrastive module is computed by contrasting a random positive pair against the set of negative pairs, which is demonstrated as follows:

$$\mathcal{L}_{\text{CL}}(H_C) = -\log \left(\frac{\exp(\text{sim}(H_C, \mathbf{S}^*(H^+))/\tau)}{\exp(\text{sim}(H_C, \mathbf{S}^*(H^+))/\tau) + \sum_{k=1}^3 \exp(\text{sim}(H_C, H^-)/\tau)} \right), \quad (7)$$

where τ is the temperature parameter that scales the similarity scores before the softmax normalization. In this work, we use the standard unscaled softmax without additional temperature tuning, and therefore set $\tau = 1$. Here, H^+ denotes the set of reason-guided content embeddings whose stance is consistent with the ground-truth label, while H^- denotes those with the opposite stance. For example, $H^+ = \{H_{r_1 \rightarrow C}, H_{r_2 \rightarrow C}, H_{r_3 \rightarrow C}\}$ when the ground-truth label is true, and $H^+ = \{H_{f_1 \rightarrow C}, H_{f_2 \rightarrow C}, H_{f_3 \rightarrow C}\}$ when the ground-truth label is false, and H^- includes the instances opposite to those in H^+ , $\mathbf{S}^*$ denotes the random sampling operator, which selects an instance from the set, and $\text{sim}(\cdot, \cdot)$ is the learnable cosine similarity evaluation measuring the closeness between two vectors. The contrastive revision effectively refines the discriminative ability of the embeddings of the textual content, ensuring that the learned representations are more aligned with the underlying truth or falsity of the news.

3.2.3. Loss function

In the base learner for fake news detection, we design a comprehensive loss $\mathcal{L}$ to train the model, including a main loss item $\mathcal{L}_{\text{Main}}$ for binary classification, a contrastive loss item $\mathcal{L}_{\text{CL}}$ to enhance the quality of the original content embedding, a reason judgment loss item $\mathcal{L}_{\text{R-Pred}}$ to align the reasons with the ground-truth label, and a reason Kullback-Leibler (KL) divergence item $\mathcal{L}_{\text{TF-KL}}$ to different the reasons supporting truthfulness and reasons supporting falseness.

As for $\mathcal{L}_{\text{Main}}$, a dynamic concatenation-based attention weight calculation module is employed to determine and normalize the weights across a set of seven main embeddings, denoted as $H^\Phi = \{H_C, H_{X_n \rightarrow C}\}$, where $n = 1, \dots, 6$.

$$\phi(H) = \text{softmax}_{H \in H^\Phi} (e(H)) = \frac{\exp(e(H))}{\sum_{H' \in H^\Phi} \exp(e(H'))} = \frac{\exp(\text{ReLU}(\bar{a}^T [W_A^T H]))}{\sum_{H' \in H^\Phi} \exp(\text{ReLU}(\bar{a}^T [W_A^T H']))}, \quad (8)$$

where W_A is the learnable matrix in the attention module, $\bar{a}$ is the learnable attention vector used to map the transformed embedding into a scalar attention score, and $e(H)$ is the concatenation-based attention coefficient of H . $\phi(H)$ can be viewed as the normalized contribution score of embedding H in the final prediction. Further, the multi-head attention can be executed for H and the average operator is selected as the pooling process. Then the main loss is shown as follows:

$$\mathcal{L}_{\text{Main}} = \text{CE} \left(\text{MLP} \left(\sum_{H \in H^\Phi} \phi(H) \cdot H \right), y \right). \quad (9)$$

where $\text{CE}(\cdot)$ denotes the standard cross-entropy loss between the predicted probabilities and the ground-truth labels.

As for $\mathcal{L}_{\text{R-Pred}}$, reasons supporting the truthfulness or falseness of news inherently align with one perspective in the classification tasks. Thus, they can also serve as auxiliary evidence for targeted retrieval of the veracity.

$$\mathcal{L}_{\text{R-Pred}} = \text{CE} \left(\text{MLP}(\text{CAT}(H_{r_n \rightarrow C})), y \right) + \text{CE} \left(\text{MLP}(\text{CAT}(H_{f_n \rightarrow C})), y \right). \quad (10)$$

As for $\mathcal{L}_{\text{TF-KL}}$, to ensure that the embeddings for reasons supporting the truthfulness and reasons supporting the falseness capture distinct views of information, we compute the KL divergence between these two categories to enhance their differentiation. Therefore, we maximize the KL divergence between these two groups by subtracting $\mathcal{L}_{\text{TF-KL}}$ in the overall objective, encouraging the model to learn more separable and stance-aware reason representations.

$$\mathcal{L}_{\text{TF-KL}} = \text{KL}(\text{CAT}(H_{r_n \rightarrow C}), \text{CAT}(H_{f_n \rightarrow C})). \quad (11)$$

The the objective loss can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\text{Main}} + \beta_1 \cdot \mathcal{L}_{\text{CL}} + \beta_2 \cdot \mathcal{L}_{\text{R-Pred}} - \beta_3 \cdot \mathcal{L}_{\text{TF-KL}}. \quad (12)$$

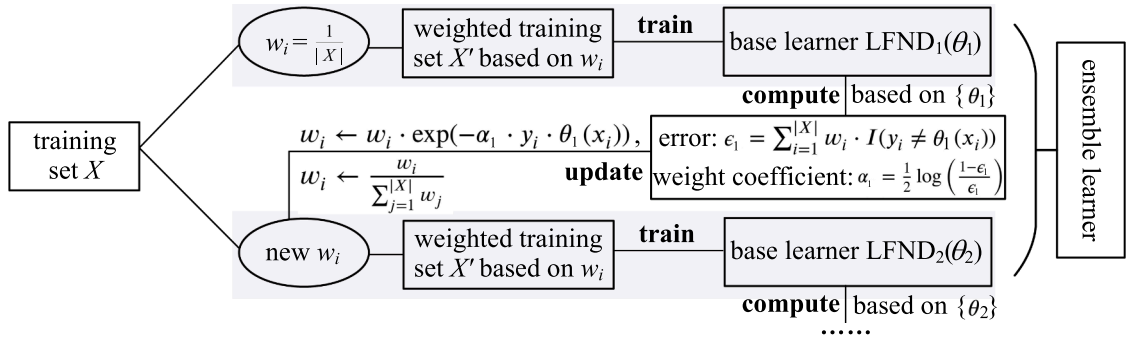
where β_1 , β_2 , and β_3 are coefficients that balance the contrastive loss $\mathcal{L}_{\text{CL}}$, the reason prediction loss $\mathcal{L}_{\text{R-Pred}}$, and the KL divergence term $\mathcal{L}_{\text{TF-KL}}$ with respect to the main ground-truth classification loss $\mathcal{L}_{\text{Main}}$.

3.3. Ensemble boosting strategy

The above section has designed an explainable base learner for fake news detection, which can be directly applied to the detection task. However, due to the common phenomenon where true news outnumbers false news in social networks, the dataset typically exhibits label imbalance. This imbalance often limits the performance of traditional small models. To address this issue, we further develop an ensemble boosting framework that combines multiple base learners and each new learner dynamically focuses on the misclassified samples in previous rounds, thereby collectively enhancing detection performance in an integrated manner. More specifically, in our adaptive boosting framework, each base learner contributes differently to the final ensemble through sequential training and error-aware weight adjustments. The first learner aims to capture the general distribution based on the average same weight loss for each sample. In contrast, the second learner focuses on correcting the errors of the first by improving the weight loss of the misclassified samples. In summary, after each boosting round, the sample distribution is updated. As a result, each base learner focuses on different aspects of the training data, which promotes error correction in subsequent stages and improves the ensemble's overall robustness.

Algorithm 1 The algorithm of LFND-AB.**Input:** Number of the base learner N , the initialized base learners θ_n in the n -th round.**Output:** The integrated model of base learners.

- 1: Initialize weights $w_i = \frac{1}{|X|}$ for all $i = 1, \dots, |X|$ of the training set X
- 2: Record the vote number $V = 0$
- 3: **for** $n = 1$ to N **do**
- 4: Fit the base learner θ_n to the training data using weights w_i //Eqs. (2)–(12)
- 5: Compute the error $\epsilon_n = \sum_{i=1}^{|X|} w_i \cdot I(y_i \neq \theta_n(x_i))$
- 6: **if** $\epsilon_n > 0.5$ **then**
- 7: Terminate the ensemble boosting strategy
- 8: **end if**
- 9: Compute $\alpha_n = \frac{1}{2} \log \left(\frac{1-\epsilon_n}{\epsilon_n} \right)$
- 10: Update weights $w_i \leftarrow w_i \cdot \exp(-\alpha_n \cdot y_i \cdot \theta_n(x_i))$
- 11: Normalize weights $w_i \leftarrow \frac{w_i}{\sum_{j=1}^{|X|} w_j}$
- 12: $V = V + 1$
- 13: **end for**
- 14: The final model is $f(\theta) = \Omega(\alpha_n \theta_n)$
- 15: **return** $f(\theta)$

**Fig. 4.** The diagram of the integrated process for ensemble learning.

After implementing the ensemble boosting strategy, the complete LFND-AB can predict fake news with higher detection performance. The algorithmic procedure of LFND-AB is illustrated in Algorithm 1. In summary, for scenarios requiring low overhead, the explainable base learner can be directly employed for detecting fake news. While in situations with imbalanced label datasets or where resources are permitted, LFND-AB offers significant advantages. These choices provide a flexible framework for fake news detection tasks in social networks. Fig. 4 demonstrates the block diagram to express the process of integrated learning.

4. Experiment

In this section, we conduct a series of experiments to confirm the effectiveness of LFND-AB. The following subsections provide detailed insights into the datasets, experimental setup, baselines, and the results obtained.

4.1. Datasets

To confirm the effectiveness of LFND-AB, this paper conducts experiments on both real-world Chinese and English datasets. Among them, the Chinese dataset is from Weibo21 (Nan et al., 2021), and the English dataset is from Gossipcop (Shu et al., 2020). Building on the methodologies of previous studies (Hu et al., 2024; Zhu et al., 2022a), the datasets are processed by removing duplicates and employing a temporal split. This approach is intended to prevent potential overestimation of performance that could result from data leakage in the SLM. The statistics for the datasets are shown in Table 1.

4.2. Experiment setup

We implement our models using pytorch and the geometric package. The following experiments are run on the server with AMD Ryzen Threadripper PRO 5995WX 64-Cores CPU and four NVIDIA RTX A6000. We use the model with the highest macro F1-score (harmonic mean of precision and recall) of the validation set during training for prediction in the testing set. For the parameters in training, the optimizer is Adam, the learning rate is 7e-5, and the number of epochs is set to 64. To reduce overfitting, we apply

Table 1
The detailed statistics of each dataset.

	Chinese			English		
	Train	Val-	Test	Train	Val-	Test
Real	2331	1172	1137	2878	1030	1024
Fake	2873	779	814	1006	244	234
Total	5204	1951	1951	3884	1274	1258

early stopping with 3 epochs, halting training if no improvement is seen in the validation set. For text encoding, we adopt `chinese-bert-wwm-ext` for Chinese and `bert-base-uncased` for English datasets with a maximum input length of 170 and a hidden size of 768. Two BERT encoders are used for the news content and rationale streams, respectively, where only the last Transformer layer is fine-tuned. The final classifier is an MLP with a structure of $768 \rightarrow 384 \rightarrow 1$ and a dropout rate of 0.2. The loss weights are set to $\beta_1 = 0.001$ for $\mathcal{L}_{CL}$, $\beta_2 = 2.0$ for $\mathcal{L}_{R_Pred}$, and $\beta_3 = 1.0$ for $\mathcal{L}_{TF_KL}$. For the boosting framework, the number of base learners is set to 2 in LFND-AB.

4.3. Baselines

To validate the effectiveness of the proposed method, we compared it with two types of baselines as follows:

4.3.1. SLM-Based methods

- **BERT** (Devlin et al., 2019) is the widely used pre-trained small language model. For Chinese and English datasets, we use `chinese-bert-wwm-ext` and `bert-base-uncased`, respectively.
- **EANN_T** (Wang et al., 2018) is a text-only variant of Event Adversarial Neural Network (EANN). EANN is a multimodal method using text and images to learn the pattern features of fake news. It innovatively employs the auxiliary task of event classification to extract the invariant features of fake news events.
- **HMCAN_T** (Qian et al., 2021) is a text-only variant of Hierarchical Multi-modal Contextual Attention Network (HMCAN). HMCAN uses BERT and ResNet to extract textual and visual features, respectively, and detects fake news by modeling both inter-modal and intra-modal relationships through a multimodal contextual attention module.
- **DualEmotion_p** (Zhang et al., 2021) is a variant of DualEmotion that uses only news content. DualEmotion uses the dual emotions of the news content (publisher) and the comments (social) to assist in the detection of fake news.
- **ENTITY DEbiasing Framework (ENDEF)** (Zhu et al., 2022a) employs causal learning techniques to eliminate entity bias, enhancing its ability to generalize across fake news datasets with distribution shifts.

4.3.2. LLM-Based methods

- **GPT-3.5-turbo** (OpenAI, 2022) is the performance of the LLM using zero-shot CoT with the commonsense perspective, following the experimental settings in the SOTA work (Hu et al., 2024).
- **SuperICL** (Xu et al., 2024) uses SLM as an in-context learning plug-in for the LLM. It enhances the testing phase by incorporating both the prediction and the confidence level of each sample directly into the prompt.
- **Adaptive Rationale Guidance network (ARG)** (Hu et al., 2024) uses SLMs to selectively harness insights derived from the rationales generated by LLMs, optimizing the process of news analysis.
- **Explanation-Guided Network (EGN)** (Phung Mai Suong et al., 2025) uses LLMs to generate common sense and sentiment explanations for news through CoT prompting, thereby enabling the detection of fake news.
- **TED** (Liu et al., 2025) uses agents supporting and questioning the veracity of news stories to simulate a human debate process for evaluating news content.

4.4. Overall performance

The fake news detection performance based on different datasets is illustrated in Table 2. It can be seen that the lightweight proposed model, i.e., explainable base learner LFND, exhibits the best detection performance among all SOTA methods. Compared to the optimal baseline ARG, LFND demonstrates an average improvement of 1.2% in the Chinese dataset and 1.1% in the English dataset. There are three reasons for the improvement: 1) The reasons supporting the truthfulness or falseness of LLM explore opposing but comprehensive perspectives, offering a broader range of specific external knowledge rather than vague generalizations. Furthermore, by requiring three reasons per perspective, the bias of the given reasons can be minimized and the robustness is enhanced. 2) The sampling based contrastive learning not only introduces randomness to avoid overfitting but also enhances the quality of textual content embeddings by encouraging proximity between positive pairs and enhancing the differentiation between negative pairs. 3) The joint loss function comprehensively considers the role of each type of embedding in the fake news detection task, ensuring the quality of all embeddings, which optimizes model training.

Then, we consider the effectiveness of the ensemble boosting strategy, and the LFND-AB is validated. It can be seen that the ensemble strategy can further enhance the detection performance. Compared to the optimal baseline ARG, LFND-AB demonstrates an average improvement of 2.4% in the Chinese dataset and 2.3% in the English dataset. The primary reason for the further improvement

Table 2

Performance metrics for different models on fake news detection. The bold values represent the best results, while underlined values denote the second-best.

Group	Model	Chinese				English			
		macF1	Acc.	$F1_{Real}$	$F1_{Fake}$	macF1	Acc.	$F1_{Real}$	$F1_{Fake}$
SLM-based Methods	BERT	0.753	0.754	0.769	0.737	0.765	0.862	0.916	0.615
	EANN _T	0.754	0.756	0.773	0.736	0.763	0.864	0.918	0.608
	HMCAN _T	0.758	0.760	0.778	0.738	0.766	0.864	0.915	0.617
	DualEmotion _p	0.761	0.763	0.784	0.738	0.766	0.868	0.920	0.611
	ENDEF	0.765	0.766	0.779	0.751	0.768	0.865	0.918	0.618
LLM-based Methods	GPT-3.5-turbo	0.725	0.734	0.774	0.676	0.702	0.813	0.884	0.519
	SuperICL	0.757	0.759	0.779	0.734	0.736	0.864	0.920	0.551
	ARG	0.784	0.786	0.804	0.764	0.790	0.878	0.926	0.653
	EGN	0.781	0.785	0.802	0.760	0.791	<u>0.889</u>	<u>0.933</u>	0.650
	TED	0.755	0.756	0.767	0.744	0.743	0.854	0.912	0.574
	LFND	<u>0.793</u>	<u>0.795</u>	<u>0.815</u>	<u>0.770</u>	<u>0.799</u>	<u>0.889</u>	<u>0.933</u>	<u>0.665</u>
	LFND-AB	0.817	0.819	0.823	0.811	0.823	0.904	0.943	0.704

Table 3

Training/testing time and GPU memory usage comparison between base learner and ensemble variant on English and Chinese datasets.

Dataset	Variant	Train Time (s/epoch)	Test Time (s/epoch)	GPU Memory (MiB)
English	Base Learner	46	11	11,478
	Ensemble (2×)	46	22	12,510
Chinese	Base Learner	63	17	11,422
	Ensemble (2×)	63	35	12,328

is that each new learner not only keeps the advantages mentioned above but also dynamically adjusts the weights of training samples based on the classification results of the previous base learner, including decreasing the weights of correctly classified samples and increasing the weights of incorrectly classified samples. Consequently, the combined capabilities of the ensemble learners result in a stronger and more accurate detection of samples. Overall, the proposed methods provide flexible options for fake news detection in different scenarios. The lightweight LFND method requires less overhead, and the LFND-AB method has better detection performance.

What's more, the results reveal distinctions between SLM-based and LLM-based baselines. While LLMs are generally considered to possess stronger semantic reasoning abilities, their performance in this task does not consistently surpass that of smaller models. Specifically, GPT-3.5-turbo performs worse than all SLM-based baselines across both datasets. This suggests that, although LLMs have access to large-scale knowledge, their performance may be hindered in specialized tasks like fake news detection, particularly when they are not fine-tuned for task-specific reasoning. Similarly, SuperICL shows improvements over some SLM-based baselines but fails to outperform SOTA SLM-based methods such as ENDEF. This may be attributed to the nature of fake news detection, which often requires understanding semantic and factual cues. While SuperICL incorporates SLM predictions and confidence scores as auxiliary input for in-context learning, this signal may be insufficient to fully represent the deeper patterns necessary for accurate detection. In contrast, ARG exhibits stronger performance, owing to its strategy of selectively utilizing LLM-generated rationales through small models, narrowing the semantic gap and avoiding overreliance on potentially noisy or hallucinated content from LLMs. This finding aligns with our design intuition in LFND, that is, rather than directly trusting LLM outputs, it is more effective to treat them as auxiliary input and apply learnable interaction mechanisms to select and integrate high-quality reasoning. Additionally, our proposed contrastive learning strategy and composite loss formulation ensure that the model does not merely memorize LLM content but learns to align relevant features and filter useful signals.

We further report the GPU memory usage, the training time, and the test time for both the single base learner and the ensemble variant. As shown in Table 3, in the English dataset, the base learner requires 46 seconds per training epoch, 11 seconds per test epoch, and uses 11,478 MiB of GPU memory. The ensemble variant with 2 base learners maintains the same training time per epoch, increases test time to 22 seconds per epoch, and uses 12,510 MiB of memory. In the Chinese dataset, the base learner takes 63 seconds for training and 17 seconds for testing per epoch, with 11,422 MiB of memory usage, while the ensemble variant takes the same 63 seconds for training, 35 seconds for testing, and consumes 12,328 MiB of memory. These results show that the ensemble strategy introduces minimal additional memory overhead, with GPU memory usage increasing by less than 10% compared to the single base learner. In terms of inference time, the increase is approximately linear with respect to the number of base learners, which confirms the efficiency and scalability of the ensemble strategy.

4.5. Statistical comparison analysis

As shown in Section 4.4, following the same seed from the SOTA method, the reported gains are modest (typically less than 3 percentage points in macF1 over ARG). To support claims of LFND's superiority and more rigorously assess whether performance im-

Table 4

Statistical comparison analysis of macF1 between LFND and ARG methods (15 paired experiments using different random seeds) in the English dataset.

Metric	Method	Mean $\pm$ Std	Significance Testing		
			paired t-statistic	p-value	95% CI
macF1	LFND	0.803 $\pm$ 0.004	5.308	<0.001***	[0.006, 0.015]
	ARG	0.792 $\pm$ 0.008			

Note: The t-statistic reflects the standardized difference between LFND and ARG over multiple runs; higher absolute values indicate stronger separation between the two methods. *p < 0.05, **p < 0.01, ***p < 0.001; CI represents 95% confidence interval for the difference (LFND - ARG).

Table 5

Detailed ranking table for MacF1 across 10 seeds. Rankings are based on performance values for each seed (1 = best, 3 = worst). Each numeric column represents a different seed set in the corresponding experimental settings.

Experimental Seed	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	Avg. Rank
LFND	1	1	1	2	1	1	1	2	2	1	1.30
ARG	2	2	2	1	2	2	2	1	1	2	1.70
BERT	3	3	3	3	3	3	3	3	3	3	3

provements are statistically significant, we further report mean $\pm$ standard deviation over 15 different seeds, and include significance testing of paired t-tests and confidence intervals. The paired t-test statistic is calculated as:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}, \quad (13)$$

where $\bar{d}$ is the mean difference between paired observations (LFND - ARG), s_d is the standard deviation of the differences, and n is the number of paired observations. A positive t-statistic indicates that LFND performs better on average, while a negative value indicates that ARG performs better.

As shown in Table 4, for the macF1 metric, LFND demonstrates significantly better performance (0.803 vs 0.792, $p < 0.001$) with a t-statistic of 5.308. The 95% confidence interval [0.006, 0.015] does not include zero, confirming the statistical significance. Overall, the mean, standard deviation, and the corresponding paired t-tests and confidence intervals over 15 different seeds indicate that LFND demonstrates stable and robust macF1 performance superiority.

To further count the significance of the performance differences among the algorithms, we conduct the non-parametric statistical test (Demšar, 2006). Specifically, we employ the Friedman test to assess overall significance. Before performing the test, we summarize the average rankings of three representative algorithms across 10 experimental seeds in Table 5. These rankings serve as the basis for subsequent statistical analysis. The tests are conducted at a significance level of $\alpha = 0.05$. The Friedman test evaluates the null hypothesis that all algorithms perform equivalently. The test statistic is computed as:

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_j R_j^2 - \frac{k(k+1)^2}{4} \right] = \frac{12N}{k(k+1)} \sum_{j=1}^k \bar{R}_j^2 - 3N(k+1), \quad (14)$$

where $N = 10$ is the number datasets using different seeds, $k = 3$ is the number of selected algorithms, and $\bar{R}_j$ denotes the average rank of the j -th algorithm. Given the average ranks $\bar{R}_1 = 1.3$, $\bar{R}_2 = 1.7$, and $\bar{R}_3 = 3.0$, we compute:

$$\chi_F^2 = \frac{12 \times 10}{3 \times 4} (1.3^2 + 1.7^2 + 3.0^2) - 3 \times 10 \times 4 = 15.8.$$

With degrees of freedom $k - 1 = 2$, the critical value from the Chi-square distribution at 95% confidence is $\chi_{0.05}^2(2) = 5.991$. Since $15.8 > 5.991$, we reject the null hypothesis and conclude that the performance differences are statistically significant. The Friedman test confirms significant differences among the representative algorithms overall.

4.6. Ablation study

In this ablation study, we first investigate the impact of the truthfulness and falseness reasons used in our model on the overall performance. Specifically, we conduct experiments using only the truthfulness reasons and using only the falseness reasons as follows:

- LFND_R uses only three truthfulness reasons instead of using all reasons.
- LFND_F uses only three falseness reasons instead of using all reasons.

Second, we further study the influence of designed components of LFND-AB on the fake news detection performance to prove their contributions. The critical modules of LFND-AB include the explainable weight evaluation module, contrastive learning nodule, joint loss function, and the ensemble boosting strategy. The variant models of LFND and LFND-AB are developed as follows:

Table 6

The detection performance evaluation of variants from LFND or LFND-AB in the English dataset.

Groups	Variants	macF1	Acc.	$F1_{\text{Real}}$	$F1_{\text{Fake}}$
Variants related to reasons integrated	LFND_R	0.769	0.868	0.921	0.617
	LFND_F	0.778	0.872	0.922	0.635
Variants related to modules introduced	LFND_W-	0.773	0.874	0.924	0.621
	LFND - $\mathcal{L}_{\text{Main}}$	0.771	0.854	0.909	0.632
	LFND - $\mathcal{L}_{\text{CL}}$	0.783	0.883	0.930	0.635
	LFND - $\mathcal{L}_{\text{R_Pred}}$	0.771	0.862	0.915	0.627
	LFND - $\mathcal{L}_{\text{TF_KL}}$	0.777	0.873	0.923	0.630
	LFND-AB_AVG	0.789	0.884	0.931	0.647
The complete methods	LFND	0.799	0.889	0.933	0.665
	LFND-AB	0.823	0.904	0.943	0.704

Table 7

The detection performance of LFND under dual-component removal in the English dataset.

Variants	macF1	Acc.	$F1_{\text{Real}}$	$F1_{\text{Fake}}$
LFND - R-F	0.762	0.870	0.922	0.603
LFND - $\mathcal{L}_{\text{R_Pred}}$ - $\mathcal{L}_{\text{CL}}$	0.764	0.870	0.922	0.605
LFND - $\mathcal{L}_{\text{R_Pred}}$ - $\mathcal{L}_{\text{TF_KL}}$	0.766	0.871	0.923	0.610
LFND - $\mathcal{L}_{\text{CL}}$ - $\mathcal{L}_{\text{TF_KL}}$	0.788	0.880	0.921	0.655

- LFND_W- uses a matrix consisting entirely of 1 to replace the explainable weight matrix $W_{C \rightarrow X_n}$ in Eq. (4).
- LFND - $\mathcal{L}_{\text{Main}}$ remove the main loss item in Eq. (9).
- LFND - $\mathcal{L}_{\text{CL}}$ remove the contrastive loss item in Eq. (7).
- LFND - $\mathcal{L}_{\text{R_Pred}}$ remove the reason prediction loss item in Eq. (10).
- LFND - $\mathcal{L}_{\text{TF_KL}}$ remove the KL divergence loss item between true and false reasons in Eq. (11).
- LFND-AB_AVG keeps the weight of each sample unchanged in the new base learners.

As can be seen from Table 6, using only the falseness reasons results in a slightly lower F1-score, suggesting that focusing solely on falseness does not provide enough information to distinguish fake news effectively. Similarly, using only the truthfulness reasons leads to a further decrease in performance, indicating that the truthfulness reasons alone are less informative and may not be sufficient for accurate fake news detection.

What's more, the results also show that removing key components from the model architecture results in varying performance reductions. Specifically, LFND_W-, which uses a matrix of 1s to replace the explainable weight matrix, shows a noticeable drop in performance. LFND_W- removes the model's ability to assign differentiated importance to each rationale and thereby disables the attention mechanism. This indicates that the attention-based weighting plays a key role in mitigating the impact of noisy or less relevant reasons. And the removal of the different loss terms also leads to a slight decrease in performance, suggesting that the loss terms contribute to model training. LFND-AB_AVG, where the weights of each sample are kept unchanged in the new base learners, performs slightly worse than the full LFND-AB, suggesting that the adaptive weight adjustment in LFND-AB contributes to better performance by focusing on the most informative samples. In conclusion, each component, including the explainable weight matrix, reason prediction, contrastive learning, and adaptive weight adjustment, plays an important role in enhancing the model's performance.

To further understand the joint effect of different components in LFND, we conduct dual-component ablation experiments by simultaneously removing two reason-related modules. The results are shown in Table 7. First, we examine the variant LFND - R-F, where all truthfulness and falseness reasons are removed. This also implicitly disables all three reason-related loss terms ($\mathcal{L}_{\text{R_Pred}}$, $\mathcal{L}_{\text{CL}}$, and $\mathcal{L}_{\text{TF_KL}}$). As expected, this variant results in the lowest overall performance, especially in detecting fake news ($F1_{\text{Fake}} = 0.603$), validating that the generated reasons by LLMs play a critical role in enhancing fake news detection. Second, when comparing different combinations of two loss terms being removed, we observe that the removal of $\mathcal{L}_{\text{R_Pred}}$ consistently leads to larger performance degradation than removing either $\mathcal{L}_{\text{CL}}$ or $\mathcal{L}_{\text{TF_KL}}$. This is reasonable since $\mathcal{L}_{\text{R_Pred}}$ is a supervised binary classification loss on reasons, directly aligned with the fake news detection objective. Therefore, it serves as a central component among the auxiliary tasks. Interestingly, the variant LFND - $\mathcal{L}_{\text{CL}}$ - $\mathcal{L}_{\text{TF_KL}}$ achieves better performance than the variant where either $\mathcal{L}_{\text{CL}}$ or $\mathcal{L}_{\text{TF_KL}}$ is individually removed (as shown in Table 6). Since both losses are designed to separate real and fake news representations in the latent space via contrastive learning and KL divergence regularization, respectively. The results imply that these two losses are mutually reinforcing. When used together, they better shape the representation space. However, when only one is considered, the joint synergy is partially broken, leading to suboptimal representation learning.

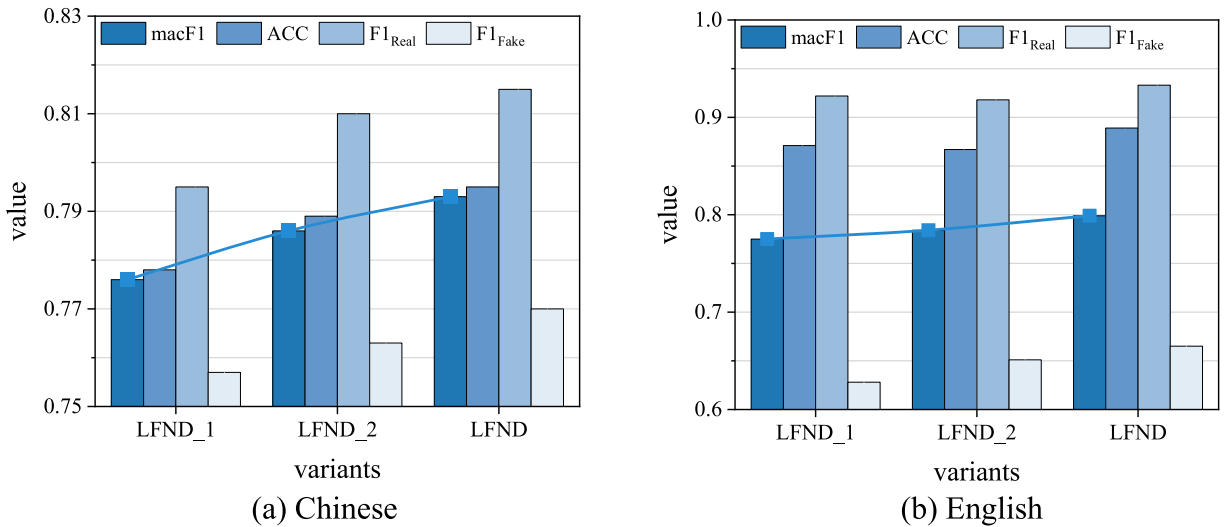


Fig. 5. Performance of LFND_1, LFND_2, and LFND on the Chinese and English datasets, showing the impact of different numbers of reason pairs.

4.7. Other experiments

4.7.1. The impact of the number of reason pairs

To explore the impact of different numbers of reason pairs on the results, we conducted experiments on both the Chinese and English datasets using one pair of reasons (LFND_1) and two pairs of reasons (LFND_2), respectively. For LFND_1, we separately use the first, second, and third sets of reasons, taking the highest performing result. Similarly, for LFND_2, we use the combinations of the first and second reasons, the second and third reasons, and the first and third reasons, again selecting the highest-performing combination.

The results in Fig. 5 show that performance improves gradually with an increasing number of reason pairs. Several reasons support this phenomenon: 1) The quality of reason pairs generated by LLM significantly contributes positively to fake news detection; 2) Our model can adaptively focus on the more valuable reasons, minimizing the impact of lower-quality reason pairs. 3) The increased number of reason pairs provides more negative sample pairs in the contrastive learning process, which has been shown to play a crucial role in enhancing model performance.

4.7.2. The impact of imbalanced data

In real-world scenarios, collecting high-quality fake news samples is generally more difficult than collecting real news. To better evaluate LFND's detection performance under such practical conditions, we conduct ablation experiments under imbalanced category data to compare the performance differences of the model when using and not using this ensemble strategy. In our original Chinese dataset, the number of real and fake news instances used for training is approximately balanced, with 2331 real news and 2873 fake news articles. To simulate increasing levels of fake news scarcity, we keep the real news portion fixed and construct two imbalanced training subsets by randomly downsampling the fake news portion:

- The 1/2 setting retains half of the original fake news (1,437 fake vs. 2331 real).
- The 1/3 setting retains only one-third of the original fake news (958 fake vs. 2331 real).

We then evaluate the performance of both the base model LFND and the boosted variant LFND-AB under the same imbalanced conditions, as shown in Table 8. From the results, we observe that under identical data imbalance conditions, LFND-AB consistently outperforms LFND. For example, in the 1/2 setting, LFND-AB improves macro-F1 by 1.57% over LFND. And in the 1/3 setting, the improvement is 1.64%. This demonstrates that the adaptive boosting strategy in LFND-AB helps the model focus on hard-to-classify instances and alleviates the performance degradation caused by label imbalance. Several factors contribute to these results. For the base learner LFND, the contrastive learning module improves the discriminative power of the embeddings by ensuring that the representations of fake and real news remain separable despite reduced training samples from the minority class. For the complete LFND-AB, the boosting strategy dynamically reweights misclassified samples during training, placing greater emphasis on challenging instances. This procedure allows the model to focus on the minority class, thus preventing the degradation of performance that often accompanies imbalanced data. Even as the amount of available fake news decreases, AB ensures that the model continues to learn effective decision boundaries.

Table 8

Performance of LFND and LFND-AB under varying levels of fake news scarcity in the Chinese dataset.

Drop Rate	0 (original)	1/3	1/2
LFND	0.7934	0.7922	0.7817
LFND-AB	0.8172	0.8086	0.7974

Table 9

Performance of reasons generated by different LLMs on the Chinese dataset.

Used LLM	ChatGPT-4.0	ChatGPT-3.5	Doubao
LFND	0.7934	0.7878	0.7863
LFND-AB	0.8172	0.8121	0.8117

4.7.3. The impact of different LLMs

To investigate the impact of using different Large Language Models to generate truthfulness and falseness reasons, we experiment with ChatGPT-3.5, ChatGPT-4.0, and Doubao⁴ on the Chinese dataset. Each LLM is employed to produce reasons supporting both the authenticity and falsity of the news samples. The results, shown in Table 9, compare the performance of LFND and LFND-AB when equipped with reasons generated by these different LLMs.

The experimental results in Table 9 indicate that ChatGPT-4.0 achieves slightly better performance than ChatGPT-3.5 and Doubao. This improvement may be attributed to ChatGPT-4.0's ability to produce more nuanced and context-rich reasons, which in turn can provide slightly more refined representations for the model. However, the overall impact on the final performance of the detection framework is not substantial. This is likely because our existing modules—such as contrastive learning and the attention mechanisms—are effective at selectively integrating and refining the information provided by the LLMs, thereby minimizing any performance discrepancies arising from the choice of LLM.

4.7.4. Case study

To demonstrate the explainable weight evaluation module in Eq. (4), we present a case study to visualize how $W_{C \rightarrow X_n}$ works, as illustrated in Table 10. The LFND model correctly identifies the given example as fake news. Among the six reasons, the third reason supporting the falseness of the news receives the highest weight, which indicates that the LFND model effectively leverages the given reasons to determine the veracity of the news content.

To investigate whether the explainable weights assigned by the attention module align with human-perceived reason quality, we conduct a small-scale human evaluation using 10 instances of news items. For each news item, five annotators are asked to score the quality of six LLM-generated reasons based on the persuasiveness of each reason in supporting or refuting the veracity of the original news content. The reason's quality scores are on a scale from 1 (least convincing quality) to 6 (most convincing quality), and each annotator is required not give repeated quality scores for six reasons to align and compare with the attention rank mechanism. And the average human-rated reason quality score of each news is computed to compare the consistency of the model attention weights. Although some individual variations exist, we consistently observe a general trend in 10 instances. Reasons assigned higher attention weights by the model tend to receive higher human quality ratings. In particular, when the model assigns significantly higher attention weights to the truthfulness reason group compared to the falseness group, the average human-rated reason quality scores for the truthfulness reasons are also notably higher.

A case study in Table 10 also reports the average human-rated reason quality score alongside the model attention weights. The results demonstrate a general alignment between the model's attention weights and human-perceived reason quality. We observe a positive correlation trend: reasons that receive higher attention weights from the model also tend to be rated as higher convincing qualities by human annotators. For example, the reason "Rumors and speculative elements..." obtains both the highest attention score (0.6876) and the highest average human-rated reason quality score (5.0). In contrast, the reason "Photographic evidence can serve as verification..." receives a relatively low attention score (0.3693), and its average human-rated reason quality score is also low (1.8). This further supports the alignment between model-assigned importance and human-perceived reasoning quality across different examples.

Further, Fig. 6 demonstrates how the base learners in the LFND-AB framework function cooperatively. During training, the first base learner is trained on the original average weighted data. The misclassified samples are assigned higher weights, and the others are assigned lower weights. Then, these weights for samples are passed to the second learner, which mainly focuses on correcting the errors. Each learner's influence on the final prediction is controlled by a normalized ensemble weight, which reflects its relative confidence. During inference, each learner makes predictions independently, and the final prediction is a weighted combination of all

⁴ <https://huggingface.co/doubao-llm>

Table 10

Case study of the explainable weights (Weight_a) assigned by the attention module ($W_{C \rightarrow X_e}$ in Eq. (4)) and the quality scores of reasons (Score_h) derived from average human evaluation. Higher attention weights indicate stronger model support for a given reason, while higher human-rated reason quality scores reflect stronger perceived support for that reason in judging news veracity.

Textual Content	Reason	Weight_a	Score_h
[Label: Fake News] Everything's fine between Sofia Richie and Scott Disick! The couple were spotted on a date just 3 weeks after cheating rumors floated around. Scott even got a bit handsy! See the pics! Scott Disick, 35, and Sofia Richie, 19, just proved that their relationship is as strong as ever. The pair met up for a smoothie date, and were all smiles as they were photographed leaving Sun Life Organics in Calabasas. At one point, Kourtney Kardashian's ex-boyfriend even pinched the model's butt while she laughed at his antics. Cute! See the PDA pictures of Scott and Sofia here. The affectionate outing comes just three weeks after the couple were enthralled in breakup rumors over an alleged cheating incident.	[Truthfulness reason] Photographic evidence can serve as verification of the event, providing visual confirmation of the couple's date and affectionate behavior.	0.3693	1.8
	[Truthfulness reason] The news mentions a specific location and recognizable public figures, adding credibility to the story.	0.6262	3.6
	[Truthfulness reason] The timeline provided (three weeks after cheating rumors) matches up with publicly known events about the couple.	0.3915	2.2
	[Falseness reason] Tabloid-style language and sensationalism could indicate that the story is exaggerated or fabricated for entertainment purposes.	0.5796	4.4
	[Falseness reason] Lack of direct quotes or statements from the individuals involved or their representatives makes the information less reliable.	0.5315	4.0
	[Falseness reason] Rumors and speculative elements, such as the alleged cheating incident, are common in gossip columns, which often blend truth with fiction.	0.6876	5.0

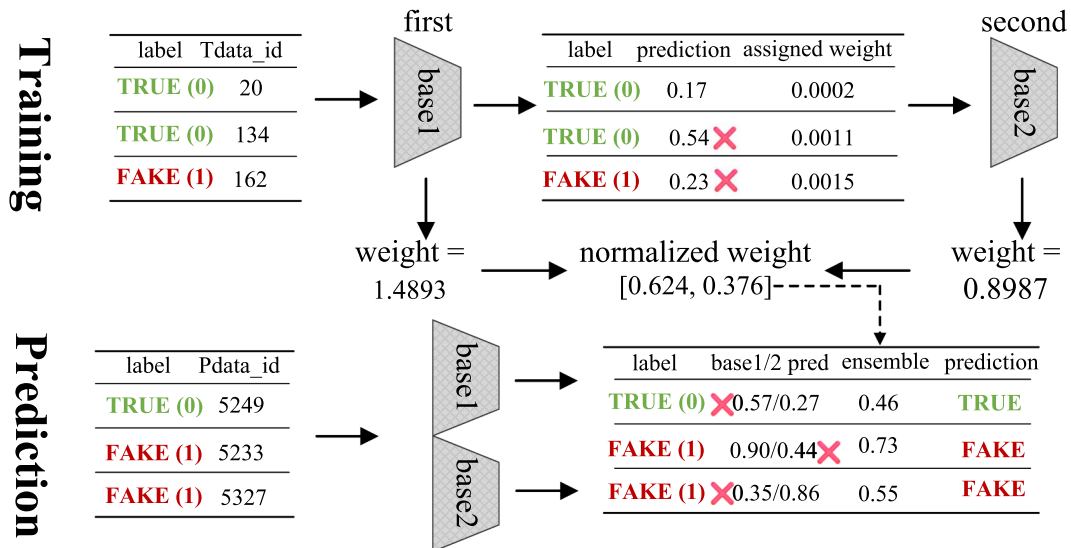


Fig. 6. An illustration of the LFND-AB adaptive boosting process. During training, each base learner is trained sequentially with updated sample weights based on previous prediction errors. The final ensemble weights are normalized and used to combine predictions from all learners during testing. This process allows each learner to focus on different aspects of the data distribution, enhancing overall model robustness.

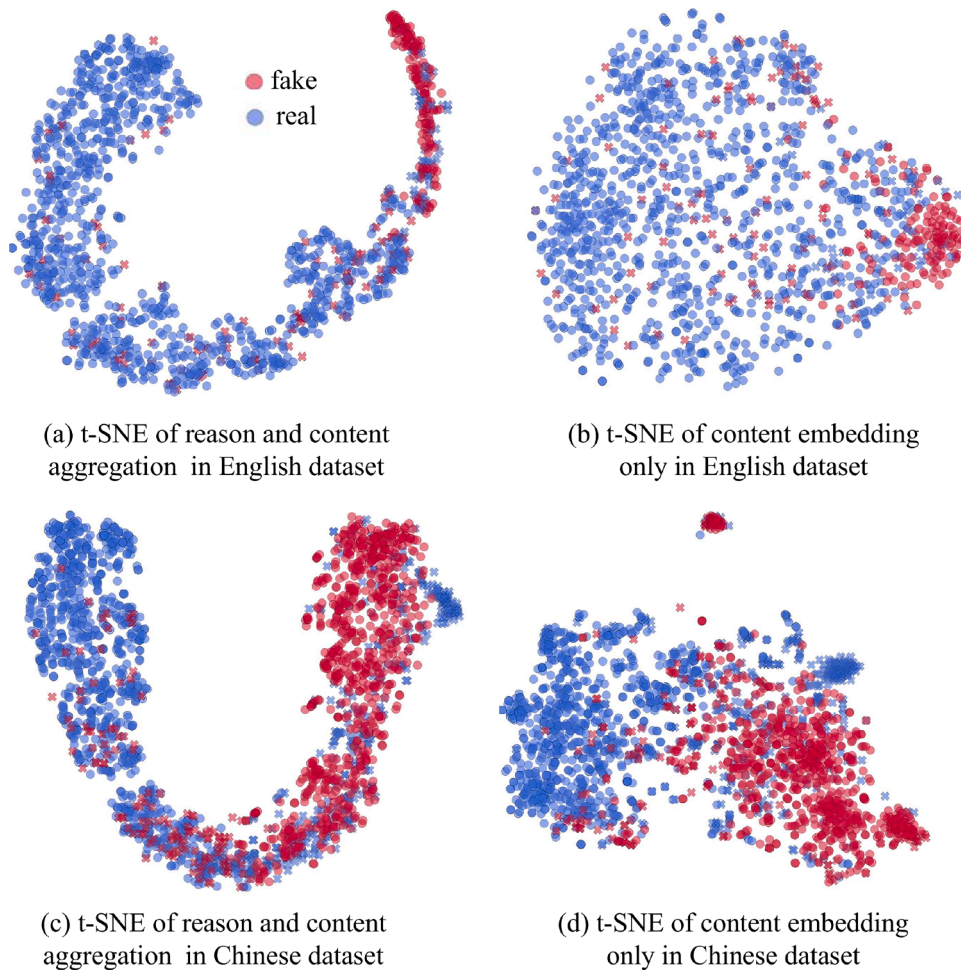


Fig. 7. t-SNE visualization of feature embeddings in English and Chinese datasets. Aggregation with reason representations (a, c) yields more separable clusters between real and fake news than using content alone (b, d), in both English and Chinese datasets. Red and blue colors represent fake and real news items. Blue crosses denote real news items that are incorrectly classified as fake and red crosses denote fake news items that are incorrectly classified as real. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

learners' outputs. This design ensures that each base learner contributes uniquely to improving the model's robustness, particularly in detecting difficult samples like fake news.

4.7.5. Feature-level visualization

To better understand how the feature space reflects the discrimination between fake and real news, t-SNE is employed to project high-dimensional features into two-dimensional space. As shown in Figs. 7 (a) and (c), the aggregated embeddings that combine content and reason representations lead to clearer separation between fake and real news, compared to using content embeddings alone in Figs. 7 (b) and (d). This observation holds in both English and Chinese datasets. These results demonstrate that integrating reasoning signals effectively enhances the discriminative power of the learned features.

We also enhance the t-SNE visualizations by explicitly marking the misclassified samples with crosses to distinguish them from correctly classified samples. In particular, we use red crosses to indicate fake news items that are incorrectly classified as real, and blue crosses to denote real news items that are incorrectly classified as fake. This more granular visualization offers a clearer picture of where the classification errors occur in the latent space and provides more actionable insight for guiding future improvements. By comparing Fig. 7 (a) which incorporates reason-content aggregation and Fig. 7 (b) which uses only content embeddings in the English dataset, we observe that the red crosses in Fig. 7 (b) are more numerous and widely scattered across the embedding space. In contrast, the misclassified samples in Fig. 7 (a) are visibly fewer and more concentrated. This indicates that the content-only embedding leads to more dispersed and confusing representations, making inaccurate classification, while the reason-content aggregation embeddings exhibit clearer separation with fewer misclassified points scattered across the opposite cluster, proving that the proposed interaction attention from reasons improves feature discriminability and reduces confusion between classes.

5. Conclusion

Given the critical role of external knowledge in fake news detection, traditional SLM-based methods either cannot learn the background information (e.g., historical facts) or require high acquisition costs. To address this, we propose an explainable LLM-assisted fake news detection method, which outperforms existing SLM-based and LLM-based methods. Additionally, considering the label imbalance caused by the facts of true news over fake news on social platforms, we introduce the LFND-AB framework, which employs an adaptive ensemble boosting strategy. By using LFND as the base learner, LFND-AB enhances the model's robustness in imbalanced data scenarios and achieves superior performance in detecting fake news, as compared to the base LFND model.

Looking ahead, several interesting directions can be further explored. First, we plan to incorporate causal reasoning mechanisms to more explicitly model the underlying cause–effect relations between LLM-generated reasons and news content, thereby strengthening the explainability and decisiveness of evidence. Second, the semantic diversity of generated reasons will be enhanced by developing diversity-oriented prompting strategies and embedding-based similarity assessment to avoid redundancy and enrich informational coverage. Third, recent studies have explored whether the learnable weights of model inference, such as attention weights or rationale scores, are aligned with human-perceived reasoning quality (Brauwers & Frasincar, 2023; Zeng et al., 2025). Incorporating human ratings or external rationale-quality supervision could further enhance the explainability of the attention mechanism.

6. Ethical considerations and limitations

The proposed LFND and LFND-AB methods have potential for integration into content management systems on social media platforms. For example, they could be deployed as automated pre-screening tools to flag potentially misleading news content during the early stages of dissemination, which helps identify high-risk posts for manual review. Additionally, the models could also be incorporated into public-facing systems, such as browser extensions, to provide users with real-time authenticity cues while consuming news. However, real-world deployment requires careful attention to several ethical concerns. First, misclassification by the model may lead to undesirable consequences. On the one hand, a high false positive rate, where large volumes of truthful news are incorrectly labeled as fake, may result in wasted human verification resources and reduced trust in the system's reliability. On the other hand, false negatives, where harmful fake news is not detected in time, could prolong public exposure to misinformation and potentially trigger widespread panic or misinformed behavior. Second, although the dual-perspective reasoning generated by LLMs improves model explainability, these generated reasons may carry cultural or linguistic biases. This may cause systematic misjudgment of news originating from specific regions or targeting certain communities. Furthermore, it is important to consider the trade-off between detection accuracy, model complexity, and computational efficiency in practical deployments. While LFND and its variant LFND-AB offer higher accuracy, they incur greater computational cost compared to lightweight models such as BERT. In real-world applications, these models can be flexibly selected based on specific performance requirements and acceptable trade-offs among accuracy, resource availability, and time constraints.

CRedit authorship contribution statement

Shu Yin: Writing – original draft, Methodology, Data curation, Conceptualization; **Yuchen Wang:** Writing – review & editing, Formal analysis; **Dongpeng Hou:** Writing – review & editing, Validation; **Wenxin An:** Writing – review & editing; **Chao Gao:** Supervision, Project administration, Funding acquisition; **Xianghua Li:** Supervision, Project administration, Funding acquisition; **Zhen Wang:** Supervision, Conceptualization, Methodology, Writing – review & editing, Funding acquisition.

Acknowledgment

This research was supported by the National Natural Science Foundation of China (Nos. U22B2036, 62471403, 62271411), the Technological Innovation Team of Shaanxi Province (No. 2025RS-CXTD-009), the International Cooperation Project of Shaanxi Province (No. 2025GH-YBXM-017).

Data availability

Data will be made available on request.

References

- Bazmi, P., Asadpour, M., & Shakery, A. (2023). Multi-view co-attention network for fake news detection by modeling topic-specific user and news source credibility. *Information Processing & Management*, 60(1).
- Bian, T., Xiao, X., Xu, T., Zhao, P., Huang, W., Rong, Y., & Huang, J. (2020). Rumor detection on social media with bi-directional graph convolutional networks. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 549–556).
- Brauwers, G., & Frasincar, F. (2023). A general survey on attention mechanisms in deep learning. *IEEE Transactions on Knowledge and Data Engineering*, 35(4), 3279–3298.
- Capuano, N., Fenza, G., Loia, V., & Nota, F. D. (2023). Content-based fake news detection with machine and deep learning: A systematic review. *Neurocomputing*, 530, 91–103.
- Chang, Y., Wang, X., Wang, J., Wu, Y., Yang, L., Zhu, K., Chen, H., Yi, X., Wang, C., Wang, Y. et al. (2024). A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*, 15(3), 1–45.
- Chen, C., & Shu, K. (2023). Combating misinformation in the age of llms: opportunities and challenges. *AI Magazine*.

- Chen, T., Li, X., Yin, H., & Zhang, J. (2018). Call attention to rumors: Deep attention based recurrent neural networks for early rumor detection. In *Trends and applications in knowledge discovery and data mining* (pp. 40–52).
- Cheung, T.-H., & Lam, K.-M. (2023). Factllama: Optimizing instruction-following language models with external knowledge for automated fact-checking. In *2023 Asia pacific signal and information processing association annual summit and conference* (pp. 846–853).
- Cui, C., & Jia, C. (2024). Propagation tree is not deep: Adaptive graph contrastive learning approach for rumor detection. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 73–81).
- Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7(Jan), 1–30.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171–4186).
- Dou, Y., Shu, K., Xia, C., Yu, P. S., & Sun, L. (2021). User preference-aware fake news detection. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 2051–2055).
- Guo, H., Zeng, W., Tang, J., & Zhao, X. (2023). Interpretable fake news detection with graph evidence. In *Proceedings of the 32nd ACM international conference on information and knowledge management* (pp. 659–668).
- Hou, D., Gao, C., Wang, Z., Li, X., & Li, X. (2024). Random full-order-coverage based rapid source localization with limited observations for large-scale networks. *IEEE Transactions on Network Science and Engineering*, 11(5), 4213–4226.
- Hu, B., Sheng, Q., Cao, J., Shi, Y., Li, Y., Wang, D., & Qi, P. (2024). Bad actor, good advisor: Exploring the role of large language models in fake news detection. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 22105–22113).
- Hu, B., Sheng, Q., Cao, J., Zhu, Y., Wang, D., Wang, Z., & Jin, Z. (2023). Learn over past, evolve for future: Forecasting temporal trends for fake news detection. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 5: Industry track)* (pp. 116–125).
- Hu, S., Zhang, C., Zou, G., Lou, Z., & Ye, Y. (2025). Deep multiview clustering by pseudo-label guided contrastive learning and dual correlation learning. *IEEE Transactions on Neural Networks and Learning Systems*, 36(2), 3646–3658.
- Jin, W., Gao, Y., Tao, T., Wang, X., Wang, N., Wu, B., & Zhao, B. (2025a). Veracity-oriented context-aware large language models-based prompting optimization for fake news detection. *International Journal of Intelligent Systems*, 2025(1), 5920142.
- Jin, W., Su, D., Tao, T., Wang, X., Wang, N., & Zhao, B. (2025b). Courtroom-FND: A multi-role fake news detection method based on argument switching-based courtroom debate. *Journal of King Saud University Computer and Information Sciences*, 37(3), 33.
- Jin, W., Wang, N., Tao, T., Jiang, M., Xing, Y., Zhao, B., Wu, H., Duan, H., & Yang, G. (2025c). A prompting multi-task learning-based veracity dissemination consistency reasoning augmentation for few-shot fake news detection. *Engineering Applications of Artificial Intelligence*, 144, 110122.
- Jin, Z., Cao, J., Zhang, Y., & Luo, J. (2016). News verification by exploiting conflicting social viewpoints in microblogs. In *Proceedings of the AAAI conference on artificial intelligence* (p. 2972–2978).
- Kuntur, S., Wróblewska, A., Paprzycki, M., & Ganzha, M. (2024). Under the influence: A survey of large language models in fake news detection. *IEEE Transactions on Artificial Intelligence*, 6(2), 458–476.
- Li, M., Peng, B., Galley, M., Gao, J., & Zhang, Z. (2024). Self-checker: Plug-and-play modules for fact-checking with large language models. In *Findings of the association for computational linguistics: NAACL 2024* (pp. 163–181).
- Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., Yang, Y., Shi, S., & Tu, Z. (2024). Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 24th conference on empirical methods in natural language processing* (pp. 17889–17904).
- Liu, Y., Liu, Y., Zhang, X., Chen, X., & Yan, R. (2025). The truth becomes clearer through debate! multi-agent systems with large language models unmask fake news. In *Proceedings of the 48th international ACM SIGIR conference on research and development in information retrieval* (pp. 504–514).
- Lu, Y.-W., Chen, C.-Y., & Li, C.-T. (2024). Dual graph networks with synthetic oversampling for imbalanced rumor detection on social media. In *Companion proceedings of the ACM web conference 2024* (p. 750–753).
- Luvembe, A. M., Li, W., Li, S., Liu, F., & Xu, G. (2023). Dual emotion based fake news detection: A deep attention-weight update approach. *Information Processing & Management*, 60(4).
- Ma, J., Gao, W., & Wong, K.-F. (2018). Rumor detection on twitter with tree-structured recursive neural networks. In *Proceedings of the 56th annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 1980–1989).
- Nan, Q., Cao, J., Zhu, Y., Wang, Y., & Li, J. (2021). Mdfend: Multi-domain fake news detection. In *Proceedings of the 30th ACM international conference on information & knowledge management* (pp. 3343–3347).
- OpenAI (2022). ChatGPT: Optimizing language models for dialogue. <https://openai.com/blog/chatgpt/>. Accessed: 2023-08-13.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A. et al. (2022). Training language models to follow instructions with human feedback. In *Advances in neural information processing systems* (pp. 27730–27744).
- Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J., & Wu, X. (2024). Unifying large language models and knowledge graphs: a roadmap. *IEEE Transactions on Knowledge and Data Engineering*, 36(7), 3580–3599.
- Phung Mai Suong, T., Huynh Nghiem, L., & Khai Tran, T. (2025). Leveraging large language models for enhanced fake news detection: A novel approach using explanation-guided networks. *Enterprise Information Systems*, (p. 2520213).
- Przybyla, P. (2020). Capturing the style of fake news. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 490–497).
- Qi, P., Cao, J., Li, X., Liu, H., Sheng, Q., Mi, X., He, Q., Lv, Y., Guo, C., & Yu, Y. (2021). Improving fake news detection by using an entity-enhanced framework to fuse diverse multimodal clues. In *Proceedings of the 29th ACM international conference on multimedia* (pp. 1212–1220).
- Qian, S., Wang, J., Hu, J., Fang, Q., & Xu, C. (2021). Hierarchical multi-modal contextual attention network for fake news detection. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 153–162).
- Sheng, Q., Cao, J., Zhang, X., Li, R., Wang, D., & Zhu, Y. (2022). Zoom out and observe: News environment perception for fake news detection. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 4543–4556).
- Sheng, Q., Zhang, X., Cao, J., & Zhong, L. (2021). Integrating pattern-and fact-based fake news detection via model preference learning. In *Proceedings of the 30th ACM international conference on information & knowledge management* (pp. 1640–1650).
- Shu, K., Mahudeswaran, D., Wang, S., Lee, D., & Liu, H. (2020). Fakenewsnet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media. *Big Data*, 8(3), 171–188.
- Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake news detection on social media: A data mining perspective. *ACM SIGKDD Explorations Newsletter*, 19(1), 22–36.
- Shu, K., Zhou, X., Wang, S., Zafarani, R., & Liu, H. (2019). The role of user profiles for fake news detection. In *Proceedings of the 2019 IEEE/ACM international conference on advances in social networks analysis and mining* (pp. 436–439).
- Strong, M., Aly, R., & Vlachos, A. (2024). Zero-shot fact verification via natural logic and large language models. In *Findings of the association for computational linguistics: EMNLP 2024* (pp. 17021–17035).
- Sun, T., Qian, Z., Dong, S., Li, P., & Zhu, Q. (2022). Rumor detection on social media with graph adversarial contrastive learning. In *Proceedings of the ACM web conference 2022* (pp. 2789–2797).
- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science (New York, N.Y.)*, 359(6380), 1146–1151.
- Wan, H., Feng, S., Tan, Z., Wang, H., Tsvetkov, Y., & Luo, M. (2024). DELL: Generating reactions and explanations for LLM-based misinformation detection. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: ACL 2024* (pp. 2637–2667).
- Wang, B., Li, X., Li, C., Fu, B., Pei, S., & Wang, S. (2024a). Why misinformation is created? detecting them by integrating intent features. In *Proceedings of the 33rd ACM international conference on information and knowledge management* (p. 2304–2314).
- Wang, B., Ma, J., Lin, H., Yang, Z., Yang, R., Tian, Y., & Chang, Y. (2024b). Explainable fake news detection with large language model via defense among competing wisdom. In *Proceedings of the ACM on web conference 2024* (pp. 2452–2463).

- Wang, J., Qian, S., Hu, J., & Hong, R. (2023). Positive unlabeled fake news detection via multi-modal masked transformer network. *IEEE Transactions on Multimedia*, 26, 234–244.
- Wang, R., Li, L., Tao, X., Wang, P., & Liu, P. (2022). Contrastive and attentive graph learning for multi-view clustering. *Information Processing & Management*, 59(4), 102967.
- Wang, Y., Ma, F., Jin, Z., Yuan, Y., Xun, G., Jha, K., Su, L., & Gao, J. (2018). Eann: Event adversarial neural networks for multi-modal fake news detection. In *Proceedings of the 24th acm sigkdd international conference on knowledge discovery & data mining* (pp. 849–857).
- Wu, J., Guo, J., & Hooi, B. (2024). Fake news in sheep's clothing: Robust fake news detection against LLM-empowered style attacks. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 3367–3378).
- Wu, J., & Hooi, B. (2023). Decor: Degree-corrected social graph refinement for fake news detection. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 2582–2593).
- Wu, J., Xu, W., Liu, Q., Wu, S., & Wang, L. (2023a). Adversarial contrastive learning for evidence-aware fake news detection with graph neural networks. *IEEE Transactions on Knowledge and Data Engineering*, 36(11), 5591–5604.
- Wu, L., Long, Y., Gao, C., Wang, Z., & Zhang, Y. (2023b). Mfir: Multimodal fusion and inconsistency reasoning for explainable fake news detection. *Information Fusion*, 100, 101944.
- Wu, L., Rao, Y., Zhang, C., Zhao, Y., & Nazir, A. (2021a). Category-controlled encoder-decoder for fake news detection. *IEEE Transactions on Knowledge and Data Engineering*, 35(2), 1242–1257.
- Wu, Y., Zhan, P., Zhang, Y., Wang, L., & Xu, Z. (2021b). Multimodal fusion with co-attention networks for fake news detection. In *Findings of the association for computational linguistics: ACL-IJCNLP 2021* (pp. 2560–2569).
- Xia, Y., Jiang, H., & Yu, S. (2025). Exploring the dynamics of group-based internet rumors propagation: A novel model from the perspective of random hypergraphs. *Information Processing & Management*, 62(1).
- Xie, B., Ma, X., Shan, X., Beheshti, A., Yang, J., Fan, H., & Wu, J. (2024). Multiknowledge and LLM-inspired heterogeneous graph neural network for fake news detection. *IEEE Transactions on Computational Social Systems*, (pp. 1–13).
- Xu, C., Xu, Y., Wang, S., Liu, Y., Zhu, C., & McAuley, J. (2024). Small models are valuable plug-ins for large language models. In *Findings of the association for computational linguistics: ACL 2024* (pp. 283–294).
- Yuan, C., Ma, Q., Zhou, W., Han, J., & Hu, S. (2019). Jointly embedding the local and global relations of heterogeneous graph for rumor detection. In *2019 IEEE International conference on data mining* (pp. 796–805).
- Zeng, Y., Ding, X., Cai, B., Liu, T., & Qin, B. (2025). Exploring large language models for effective rumor detection on social media. In *Proceedings of the 2025 conference of the nations of the americas chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 2537–2552).
- Zhang, H., Liu, X., Yang, Q., Yang, Y., Qi, F., Qian, S., & Xu, C. (2024). T3RD: Test-time training for rumor detection on social media. In *Proceedings of the ACM on web conference 2024* (pp. 2407–2416).
- Zhang, X., Cao, J., Li, X., Sheng, Q., Zhong, L., & Shu, K. (2021). Mining dual emotion for fake news detection. In *Proceedings of the web conference 2021* (p. 3465–3476).
- Zhou, X., & Zafarani, R. (2020). A survey of fake news: Fundamental theories, detection methods, and opportunities. *ACM Computing Surveys*, 53(5), 1–40.
- Zhu, Y., Sheng, Q., Cao, J., Li, S., Wang, D., & Zhuang, F. (2022a). Generalizing to the future: Mitigating entity bias in fake news detection. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval* (pp. 2120–2125).
- Zhu, Y., Sheng, Q., Cao, J., Nan, Q., Shu, K., Wu, M., Wang, J., & Zhuang, F. (2022b). Memory-guided multi-view multi-domain fake news detection. *IEEE Transactions on Knowledge and Data Engineering*, 35(7), 7178–7191.